

AD EXP-3

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Batch: S-5

Code:

JAVA CODE:

HomeActivity.java:

```
package com.example.loginapp;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;

public class HomeActivity extends AppCompatActivity {

    Button btnLogin, btnRegister;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_home);

        btnLogin = findViewById(R.id.btnLogin);
        btnRegister = findViewById(R.id.btnRegister);

        btnLogin.setOnClickListener(v ->
            startActivity(new Intent(this, LoginActivity.class)));

        btnRegister.setOnClickListener(v ->
            startActivity(new Intent(this, RegisterActivity.class)));
    }
}
```

RegisterActivity.java:

```
package com.example.loginapp;

import android.os.Bundle;
import android.widget.*;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;

public class RegisterActivity extends AppCompatActivity {

    EditText etUsername, etEmail, etPassword;
```

```
Button btnRegister;  
DatabaseHelper db;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_register);
```

```
    db = new DatabaseHelper(this);
```

```
    etUsername = findViewById(R.id.etUsername);  
    etEmail = findViewById(R.id.etEmail);  
    etPassword = findViewById(R.id.etPassword);  
    btnRegister = findViewById(R.id.btnRegister);
```

```
    btnRegister.setOnClickListener(v -> {
```

```
        String username = etUsername.getText().toString();  
        String email = etEmail.getText().toString();  
        String password = etPassword.getText().toString();
```

```
        boolean inserted = db.insertData(username, email, password);
```

```
        if(inserted) {
```

```
            Intent intent = new Intent(RegisterActivity.this, WelcomeActivity.class);  
            intent.putExtra("username", username);  
            startActivity(intent);
```

```
        } else {
```

```
            Toast.makeText(this, "Registration Failed", Toast.LENGTH_SHORT).show();
```

```
        }
```

```
    }); } }
```

LoginActivity.java:

```
package com.example.loginapp;
```

```
import android.os.Bundle;
```

```
import android.widget.*;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.content.Intent;
```

```
public class LoginActivity extends AppCompatActivity {
```

```
    EditText etEmail, etPassword;
```

```
    Button btnLogin;
```

```
    DatabaseHelper db;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_login);

db = new DatabaseHelper(this);

etEmail = findViewById(R.id.etLoginEmail);
etPassword = findViewById(R.id.etLoginPassword);
btnLogin = findViewById(R.id.btnLogin);

btnLogin.setOnClickListener(v -> {

    String email = etEmail.getText().toString();
    String password = etPassword.getText().toString();

    String username = db.checkUser(email, password);

    if(username != null) {

        Intent intent = new Intent(LoginActivity.this, WelcomeActivity.class);
        intent.putExtra("username", username);
        startActivity(intent);

    } else {
        Toast.makeText(this, "Invalid Credentials", Toast.LENGTH_SHORT).show();
    }
});
}
}

```

DatabaseHelper.java:

```

package com.example.loginapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "UserDB.db";
    private static final String TABLE_NAME = "users";

    public DatabaseHelper(Context context) {
        super(context, DATABASE_NAME, null, 1);
    }

    @Override
    public void onCreate(SQLiteDatabase db) {

```

```

        db.execSQL("CREATE TABLE users (ID INTEGER PRIMARY KEY
AUTOINCREMENT, USERNAME TEXT, EMAIL TEXT, PASSWORD TEXT)");
    }

    @Override
    public void onUpgrade(SQLiteDatabase db, int i, int i1) {
        db.execSQL("DROP TABLE IF EXISTS users");
        onCreate(db);
    }

    public boolean insertData(String username, String email, String password) {
        SQLiteDatabase db = this.getWritableDatabase();
        ContentValues values = new ContentValues();
        values.put("USERNAME", username);
        values.put("EMAIL", email);
        values.put("PASSWORD", password);
        long result = db.insert(TABLE_NAME, null, values);
        return result != -1;
    }

    public String checkUser(String email, String password) {
        SQLiteDatabase db = this.getReadableDatabase();
        Cursor cursor = db.rawQuery(
            "SELECT USERNAME FROM users WHERE EMAIL=? AND PASSWORD=?",
            new String[]{email, password});

        if(cursor.moveToFirst()) {
            return cursor.getString(0); // return username
        }

        return null; // invalid login
    }
}

```

XML CODE:

activity_home.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:gravity="center"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="200dp"
        android:layout_height="wrap_content"

```

```

        android:text="Sign In"/>

<Button
    android:id="@+id/btnRegister"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:text="Sign Up"
    android:layout_marginTop="20dp"/>

</LinearLayout>

```

activity_register.xml:

```

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:padding="20dp"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <EditText
        android:id="@+id/etUsername"
        android:hint="Username"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etEmail"
        android:hint="Email"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <EditText
        android:id="@+id/etPassword"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <Button
        android:id="@+id/btnRegister"
        android:text="Register"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>

```

activity_login.xml:

```

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"

```

```
android:padding="20dp"  
android:layout_width="match_parent"  
android:layout_height="match_parent">
```

```
<EditText  
    android:id="@+id/etLoginEmail"  
    android:hint="Email"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

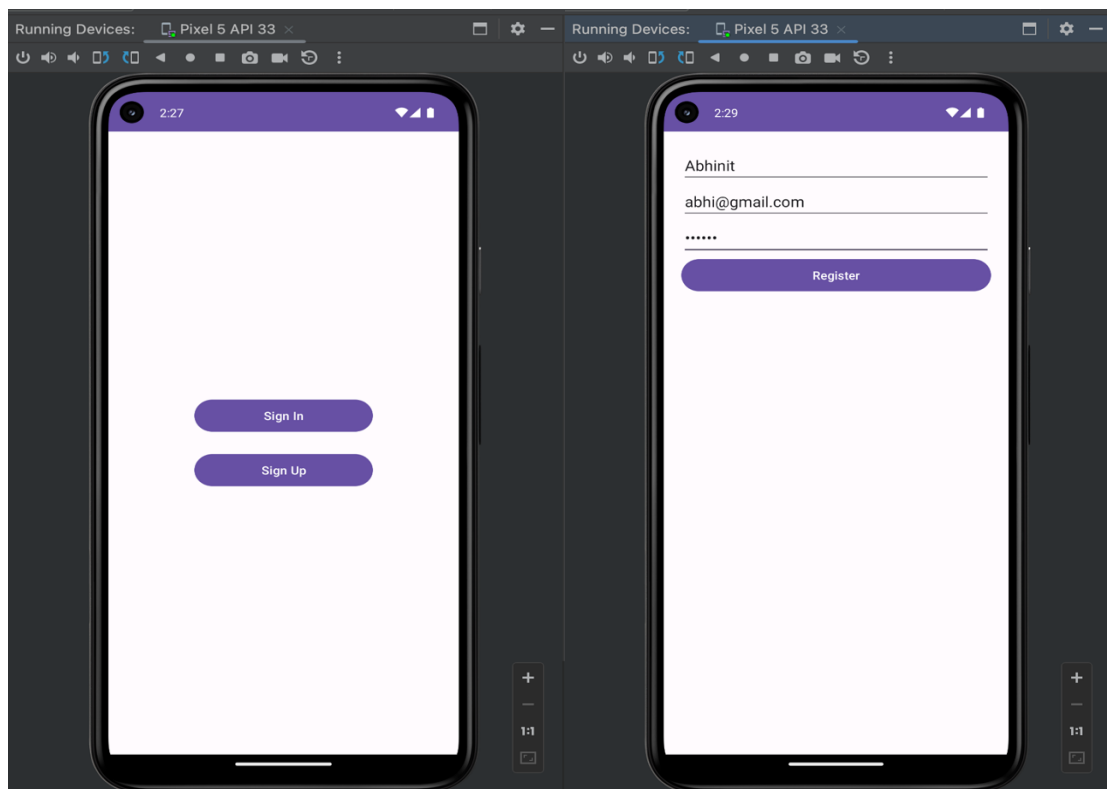
```
<EditText  
    android:id="@+id/etLoginPassword"  
    android:hint="Password"  
    android:inputType="textPassword"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

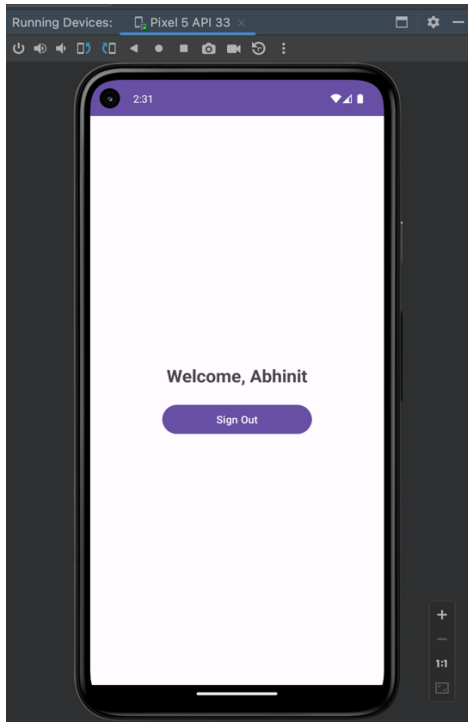
```
<Button  
    android:id="@+id/btnLogin"  
    android:text="Login"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"/>
```

```
</LinearLayout>
```

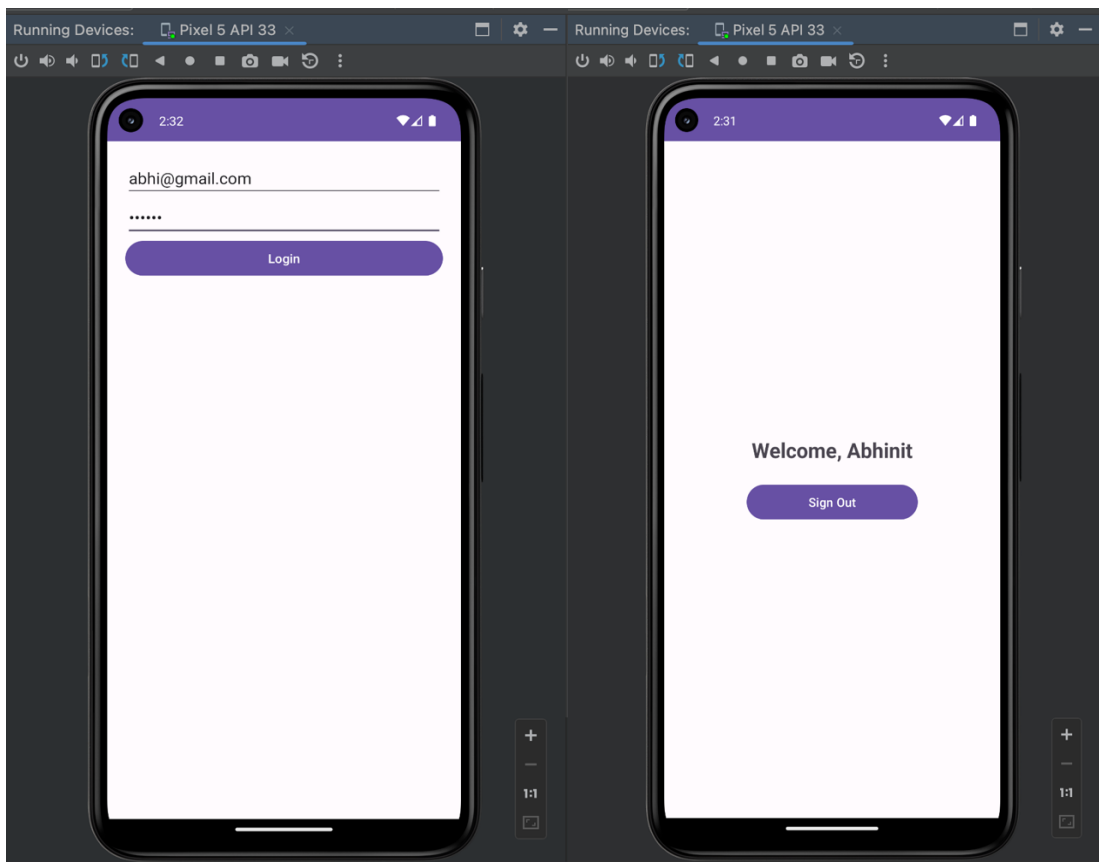
O/P:

1.) Signup





2.) Login



3.) Signout

